

MASTER BLUEPRINT v3.0 – PART 1

INDIA AI-FIRST REAL ESTATE PLATFORM

Product Vision, Business Rules, User Roles, Authentication & Platform Foundation

Version: 3.0

Status: Final Master Blueprint

Target Market:

- Gujarat (Phase 1)
- Pan India (Future)

Platform Type:

- Real Estate Marketplace
 - Real Estate SaaS Platform
 - Property Discovery Platform
 - Lead Generation Platform
 - Project Marketing Platform
-

1. PROJECT VISION

Build India's most trusted AI-First Real Estate Platform where users can:

- Buy Properties
- Sell Properties
- Rent Properties
- Lease Properties
- Discover New Projects
- Connect with Brokers
- Connect with Builders
- Connect with Agencies
- Manage Leads

- Promote Listings
- Purchase Advertising

Core Goal:

User must discover relevant property within 60 seconds.

2. CORE PLATFORM PRINCIPLES

Rule 1 – Mobile First

Entire platform optimized for mobile devices.

Rule 2 – Performance First

Homepage load:

< 2 seconds

Search:

< 1 second

Property Detail:

< 500ms

Rule 3 – Security First

Mandatory:

- JWT Authentication
- OTP Verification
- Row Level Security
- RBAC
- Audit Logs
- Rate Limiting

Rule 4 – Database Driven

Business rules should be controlled through database.

Avoid hardcoded business logic.

Rule 5 – Multi Tenant Ready

Future support:

- Agencies
 - Builders
 - Franchise Networks
 - Enterprise Accounts
-

3. USER TYPES

Public Users

Visitors can:

- Browse Properties
- Search Properties
- View Projects
- Read Blogs
- Read FAQs

Without Login.

Registered User Types

1. Buyer / Seller

Single account type.

User can switch modes.

Buyer Features:

- Search Properties
- Save Properties
- Compare Properties
- Contact Owners
- Contact Brokers
- Contact Builders
- Book Visits
- Receive Alerts

Seller Features:

- Create Listings
 - Edit Listings
 - Manage Listings
 - View Leads
 - Renew Plans
 - Purchase Promotions
-

2. Broker

Professional Property Broker.

Features:

- Multiple Listings
- Lead CRM
- Lead Tracking
- Property Promotion
- Analytics
- Team Expansion (Future)
- Subscription Plans

Requirements:

- KYC
 - Mobile Verification
 - Email Verification
-

3. Builder

Project Developer.

Features:

- Create Projects
- Manage Towers

- Manage Phases
- Upload Floor Plans
- Upload Amenities
- Manage Inventory
- Receive Leads
- Promote Projects

Requirements:

- Business Verification
 - RERA Details
 - KYC
-

4. Agency

Enterprise Account.

Features:

- Team Management
- Agent Accounts
- Lead Assignment
- Team Analytics
- Multiple Brands
- Multiple Projects
- Multiple Brokers
- Lead Distribution Engine

Requirements:

- Company Verification
 - KYC
 - Business Documents
-

4. BACKEND ROLES

Not publicly selectable.

Admin

Features:

- Property Moderation
 - KYC Verification
 - User Management
 - Support Tickets
 - Reports
 - Content Moderation
-

Super Admin

Complete platform control.

Features:

- User Control
 - Role Management
 - Feature Management
 - Revenue Management
 - Subscription Management
 - Advertisement Management
 - Platform Settings
 - Analytics Dashboard
-

5. REGISTRATION SYSTEM

Supported Methods:

- Mobile OTP
- Email OTP
- Google Login

Future:

- Apple Login
 - Facebook Login
-

6. LOGIN RULES

Guest users may:

- Search Properties
- View Listings
- View Projects

Login Required For:

- Contact Owner
 - WhatsApp Owner
 - Save Property
 - Compare Property
 - Book Visit
 - Post Property
 - Purchase Plans
 - Purchase Ads
-

7. ONBOARDING FLOW

Step 1

Select Account Type:

- Buyer / Seller
- Broker
- Builder
- Agency

Step 2

Verify Mobile

Step 3

Verify Email

Step 4

Complete Profile

Step 5

Dashboard Access

8. PROFILE SYSTEM

Store:

- Name
- Mobile
- Email
- Profile Photo
- City
- State
- Country

Optional:

- Bio
 - Website
 - Social Links
-

9. PROPERTY CATEGORIES

Residential Sale

Residential Rent

Commercial Sale

Commercial Rent

Industrial

Agricultural

Plots

Farm Houses

PG

Hostels

Co-Living

Student Housing

Offices

Shops

Warehouses

10. PROPERTY STATUS SYSTEM

Draft

Pending Review

Approved

Rejected

Active

Sold

Rented

Expired

Archived

Hidden

Rules:

New Listing:

Pending Review

Approval Required:

Admin

Approved:

Active

Expired Plan:

Hidden

Renewed Plan:

Active Again

11. LISTING RULES

Required:

- Title
- Price
- Property Type
- Location
- Area
- Contact Information
- Minimum 3 Images

Optional:

- Video
- Documents
- Floor Plans
- Virtual Tour

12. OWNER TRANSPARENCY RULE

Every listing must clearly show:

- Owner
- Broker
- Builder
- Agency

Users must always know who posted the property.

No hidden ownership.

13. GEO VERIFICATION

Optional Feature.

When enabled:

- User must be near property location
- GPS validation required

Purpose:

Reduce fake listings.

14. RERA RULES

Store:

- RERA Number
- RERA Status
- Verification Badge

Display publicly.

15. VISIT BOOKING SYSTEM

User selects:

- Date
- Time

Property Owner receives request.

Owner can:

- Approve
- Reject
- Reschedule

Visit status tracked.

16. SAVED PROPERTY SYSTEM

Users can:

- Save Properties
 - Create Favorites
 - Compare Properties
 - Create Shortlists
-

17. ALERT SYSTEM

Users receive:

- New Matching Properties
- Price Drop Alerts
- Visit Updates
- Lead Updates
- Plan Expiry Alerts

Channels:

- Email
 - SMS
 - WhatsApp
 - Push Notifications
-

18. PLATFORM SUCCESS METRICS

Primary Goals:

- Property Discovery < 60 Seconds
- Search Response < 1 Second
- Mobile First Experience
- High Lead Conversion
- Low Fraud Listings
- Scalable SaaS Foundation

PART 1 COMPLETE

MASTER BLUEPRINT v3.0 – PART 2

PROPERTY SYSTEM, PROJECT SYSTEM, SEARCH ENGINE, AI FEATURES, LEAD CRM & MESSAGING ARCHITECTURE

Version: 3.0

Status: Final Master Blueprint

19. PROPERTY MANAGEMENT SYSTEM

The Property Module is the core engine of the platform.

Supported By:

- Buyer/Seller
 - Broker
 - Builder
 - Agency
-

PROPERTY CREATION FLOW

Step 1:

Select Property Type

Step 2:

Select Listing Type

Step 3:

Enter Property Details

Step 4:

Upload Images

Step 5:

Upload Documents

Step 6:

Map Location

Step 7:

Preview

Step 8:

Submit

Step 9:

Admin Review

Step 10:

Publish

20. PROPERTY TYPES

Residential Sale

Residential Rent

Commercial Sale

Commercial Rent

Industrial

Agricultural

Plots

Farm Houses

PG

Hostels

Student Housing

Co-Living

Offices

Shops

Warehouses

Mixed Use Property

Future Expansion:

Any custom property category.

21. PROPERTY DATA STRUCTURE

Basic Information:

- Title
- Description
- Property Type
- Listing Type

Specifications:

- BHK
- Bathrooms
- Balconies
- Furnishing
- Floor Number
- Total Floors
- Property Age

Area Information:

- Super Built-Up Area
- Carpet Area
- Plot Area

Area Units:

- Sq.Ft.
- Sq.Yd.
- Guz
- Bigha
- Acre
- Hectare

Pricing:

- Price

- Rent
- Deposit
- Maintenance Charges
- Price Per Sq.Ft.

Ownership:

- Owner
- Broker
- Builder
- Agency

22. PROPERTY MEDIA SYSTEM

Supported:

Images

Videos

Floor Plans

Documents

Brochures

Virtual Tours

IMAGE RULES

Auto Compression

Auto Resize

WebP Generation

AVIF Generation

Watermarking

Virus Scan

Duplicate Detection

Quality Validation

Reject:

- Blurry Images
- Fake Images
- Dark Images
- Duplicate Images

Minimum Images:

3

Maximum Images:

Plan Controlled

23. PROPERTY LOCATION SYSTEM

Store:

- Country
- State
- District
- City
- Locality
- Society
- Address
- Pincode

GPS:

- Latitude
- Longitude

Map Providers:

Primary:

Ola Maps

Fallback:

Google Maps

24. PROPERTY AMENITIES SYSTEM

Examples:

- Lift
- Parking
- CCTV
- Security
- Club House
- Gym
- Garden
- Swimming Pool
- Children's Play Area
- Power Backup
- EV Charging

Amenities must be searchable.

25. PROPERTY VASTU SYSTEM

Store:

- Facing
- Entrance Direction
- Kitchen Direction
- Bedroom Direction

Searchable Filters.

26. PROPERTY STATUS TRACKING

Status History:

Draft

Pending Review

Approved

Rejected

Published

Sold

Rented

Expired

Archived

Hidden

All status changes logged.

27. PROPERTY COMPARISON SYSTEM

Users can compare:

- Price
- Area
- Amenities
- Location
- Builder
- BHK
- Furnishing

Comparison limit controlled by plan.

28. SAVED SEARCH SYSTEM

Users can save searches.

Examples:

2 BHK Rajkot Under 50 Lakh

Commercial Office Ahmedabad

Plot Near Ring Road

ALERTS

When matching property arrives:

- Email
 - SMS
 - WhatsApp
 - Push Notification
-

29. PRICE DROP SYSTEM

If property price reduces:

Notify:

- Interested Users
- Saved Users
- Followers

Channels:

- Push
 - Email
 - WhatsApp
-

30. SEARCH ENGINE ARCHITECTURE

Search Methods:

1. Text Search
2. AI Search
3. Voice Search
4. Map Search
5. Polygon Search
6. Nearby Search

SEARCH SCALING ROADMAP

Phase 1:

PostgreSQL + pg_trgm + PostGIS

Phase 2:

Redis Search Cache

Phase 3:

OpenSearch / Elasticsearch

Phase 4:

Dedicated Search Cluster

FILTERS

Location

Property Type

Listing Type

Budget

BHK

Area

Area Unit

Amenities

Possession Status

Builder

Furnishing

Property Age

RERA

Vastu

Society Rules

Geo Verified

31. AI SEARCH SYSTEM

Natural Language Search.

Examples:

"2 BHK Rajkot under 50 lakh"

"Office near SG Highway"

"Plot under 25 lakh in Ahmedabad"

System converts sentence into filters.

AI SEARCH RULES

Must understand:

Price

Location

Property Type

Area

Amenities

Builder Names

Local Language Terms

Examples:

Ghar → Residential

Makan → Residential

Shop → Commercial

Plot → Plot

Sasta → Low Price

Premium → Luxury

AI USAGE CONTROL

Track:

AI Requests

AI Tokens

AI Costs

Plan Limits:

Free

Premium

Enterprise

Emergency Kill Switch Supported

AI Usage Fully Auditable

32. VOICE SEARCH SYSTEM

Languages:

English

Hindi

Gujarati

Process:

Voice

↓

Speech To Text

↓

Filter Extraction

↓

Search Results

33. MAP SEARCH SYSTEM

Users can:

- Search by Radius
- Search by Area
- Search by Polygon
- Search Near Landmark

Examples:

Properties within 5km

Properties around airport

Properties inside drawn area

34. PROJECT MANAGEMENT SYSTEM

Builder Projects Structure:

Project

↓

Phase

↓

Tower

↓

Unit

35. PROJECT INFORMATION

Store:

- Project Name

- Builder Name
 - Description
 - RERA Number
 - Launch Date
 - Possession Date
-

36. PROJECT MEDIA

Store:

- Images
 - Floor Plans
 - Brochures
 - Videos
-

37. PROJECT INVENTORY SYSTEM

Unit Types:

1 BHK

2 BHK

3 BHK

4 BHK

Villa

Office

Shop

Warehouse

Custom Units

Track:

Availability

Booked

Reserved

Sold

38. CONSTRUCTION UPDATE SYSTEM

Builders can post:

- Progress Updates
- Images
- Videos
- Milestones

Users can follow project updates.

39. LEAD MANAGEMENT SYSTEM

Lead Sources:

Call

WhatsApp

Form

Visit Request

Project Inquiry

Advertisement

Referral

Organic Search

40. LEAD STATUS PIPELINE

New

Contacted

Interested

Visit Scheduled

Negotiation

Closed

Lost

Spam

All changes logged.

41. LEAD ROUTING ENGINE

Direct Property Match

↓

Agency Routing

↓

Round Robin Assignment

↓

Manual Assignment

↓

Admin Queue

AGENCY LEAD ROUTING

Lead 1 → Agent A

Lead 2 → Agent B

Lead 3 → Agent C

Lead 4 → Agent A

Repeat.

42. LEAD CRM

Store:

- Lead Notes
- Follow Ups
- Reminders
- Communication History

Track:

Calls

WhatsApp

Emails

Meetings

Visits

43. VISIT MANAGEMENT SYSTEM

Request Visit

↓

Owner Approval

↓

Schedule

↓

Reminder

↓

Visit Complete

↓

Feedback

44. REAL-TIME CHAT SYSTEM

Supported Users:

Buyer

Seller

Broker

Builder

Agency

CHAT FEATURES

One To One Chat

Property Linked Chat

Media Sharing

Document Sharing

Unread Counter

Typing Indicator (Future)

Read Receipts

Online Status (Future)

45. CONVERSATION STRUCTURE

Conversation

↓

Participants

↓

Messages

↓

Read Status

↓

Notifications

46. NOTIFICATION TRIGGERS

Lead Generated

Visit Booked

Message Received

Price Drop

Property Approved

Plan Expiry

Project Update

Payment Success

47. FRAUD DETECTION

Detect:

Duplicate Listings

Duplicate Numbers

Fake Images

Spam Leads

Repeated OTP Abuse

Suspicious Pricing

Flag For Review.

48. ANALYTICS TRACKING EVENTS

Property View

Project View

Lead Generated

WhatsApp Click

Call Click

Visit Request

Save Property

Compare Property

Search Query

Ad Click

PART 2 COMPLETE

MASTER BLUEPRINT v3.0 – PART 3

SUBSCRIPTIONS, PAYMENTS, REVENUE SYSTEM, ADVERTISEMENT SYSTEM, KYC, VERIFICATION, ANALYTICS, ADMIN & SUPER ADMIN ARCHITECTURE

Version: 3.0

Status: Final Master Blueprint

49. REVENUE ARCHITECTURE

Platform Revenue Sources:

1. Subscription Plans
2. Featured Listings
3. Homepage Ads
4. Category Ads
5. Builder Promotions
6. Agency Promotions
7. Premium Visibility
8. Future Lead Marketplace
9. Future API Services

All revenue must be trackable.

50. SUBSCRIPTION SYSTEM

Supported Roles:

Buyer/Seller

Broker

Builder

Agency

PLAN TYPES

FREE PLAN

Features:

- Limited Listings
 - Basic Visibility
 - Limited Leads
 - No Featured Listing
-

BASIC PLAN

Features:

- More Listings
 - Lead Access
 - Basic Analytics
 - Priority Support
-

PREMIUM PLAN

Features:

- Higher Listing Limits
 - Featured Listings
 - Better Visibility
 - Advanced Analytics
-

ENTERPRISE PLAN

Features:

- Team Accounts
- Unlimited Listings
- Lead Routing
- Agency Features
- Dedicated Support

51. PLAN CONTROLS

Control:

- Listing Limits
- Lead Limits
- Featured Limits
- Image Limits
- Video Limits
- Team Members
- Ad Credits
- Analytics Access
- API Access (Future)

All plan limits configurable.

52. PLAN EXPIRY SYSTEM

Notifications:

30 Days

15 Days

7 Days

3 Days

1 Day

Expired

EXPIRED PLAN RULES

Hide Listings

Disable New Uploads

Retain Data

Allow Renewal

Keep Leads Accessible

53. SUBSCRIPTION UPGRADE SYSTEM

Supported:

Upgrade

Downgrade

Renew

Cancel

Pause (Future)

Resume (Future)

54. PAYMENT SYSTEM

Gateway:

Razorpay

Future:

PhonePe

Paytm

Stripe

PAYMENT METHODS

UPI

Cards

Net Banking

Wallets

EMI

Corporate Payments

55. PAYMENT FLOW

Select Plan



Checkout



Razorpay



Webhook



Verification



Invoice



Subscription Activated

56. PAYMENT STATUSES

Pending

Processing

Success

Failed

Refunded

Cancelled

Disputed

57. WEBHOOK SYSTEM

Mandatory Webhooks:

Payment Success

Payment Failure

Refund

Subscription Renewal

Subscription Expiry

All events logged.

WEBHOOK RETRY SYSTEM

Retry Attempts:

1 Minute

5 Minutes

15 Minutes

1 Hour

Dead Letter Queue Supported

All Retry Attempts Logged

58. PAYMENT RECONCILIATION SYSTEM

Track:

Gateway Payments

Failed Payments

Refunds

Disputes

Daily Reconciliation Required

59. INVOICE SYSTEM

Generate:

GST Invoice

Payment Receipt

Refund Receipt

Credit Note

Store:

PDF Copy

Invoice Number

Tax Information

Payment Reference

60. PROPERTY ENQUIRY SYSTEM

Users can:

- Contact Owner
- Contact Broker
- Contact Builder
- WhatsApp
- Call
- Send Enquiry
- Book Visit

Lead created automatically in CRM.

All enquiries tracked.

Statuses:

New

Contacted

Interested

Visit Scheduled

Negotiation

Closed

Lost

Spam

61. ADVERTISEMENT SYSTEM

Supported Advertisers:

Buyer/Seller

Broker

Builder

Agency

External Advertisers (Future)

62. ADVERTISEMENT TYPES

Featured Listing

Homepage Banner

Search Result Promotion

Category Banner

City Banner

Project Promotion

Builder Promotion

Agency Promotion

63. AD PACKAGE SYSTEM

Package Controls:

Duration

Visibility

Impressions

Clicks

Geographic Reach

Category Reach

Priority Score

64. FEATURED LISTING SYSTEM

Featured Properties appear:

Homepage

Search Results

Category Pages

City Pages

Featured placement controlled by package.

65. AD CAMPAIGN SYSTEM

Campaign Fields:

Name

Start Date

End Date

Budget

Target Area

Target City

Target Category

Status

66. AD ANALYTICS

Track:

Impressions

Clicks

CTR

Leads Generated

Cost Per Lead

Revenue

Conversions

67. KYC SYSTEM

Applicable To:

Broker

Builder

Agency

Optional For Seller

KYC DOCUMENTS

PAN

Aadhaar

GST

Business Registration

RERA Certificate

Address Proof

Company Documents

68. KYC STATUSES

Draft

Submitted

Under Review

Approved

Rejected

Suspended

69. BUSINESS VERIFICATION

Verify:

Company

Broker

Builder

Agency

Verification Badge Displayed Publicly.

70. RERA VERIFICATION SYSTEM

Store:

RERA Number

Issue Date

Expiry Date

Verification Status

Display Badge

71. DOCUMENT MANAGEMENT SYSTEM

Store:

KYC Files

Ownership Documents

Brochures

Floor Plans

Invoices

Certificates

Contracts

Version History Supported.

72. NOTIFICATION ARCHITECTURE

Channels:

Email

SMS

WhatsApp

Push Notification

In-App Notification

73. NOTIFICATION EVENTS

User Registration

OTP Verification

Property Approval

Property Rejection

Lead Generated

Visit Booked

Visit Approved

Message Received

Price Drop

Payment Success

Refund

Plan Expiry

Project Update

Admin Announcement

74. TEMPLATE MANAGEMENT

Templates For:

Email

SMS

WhatsApp

Push

All templates managed from Admin Panel.

75. ANALYTICS ARCHITECTURE

Primary:

PostHog

Secondary:

Google Analytics

76. PLATFORM ANALYTICS

Track:

Users

Properties

Projects

Leads

Revenue

Ads

Subscriptions

Cities

Categories

77. PROPERTY ANALYTICS

Views

Unique Views

Clicks

Lead Generation

WhatsApp Clicks

Call Clicks

Visit Requests

Conversions

78. PROJECT ANALYTICS

Project Views

Brochure Downloads

Lead Generation

Booking Requests

Conversions

79. USER ANALYTICS

Registrations

Logins

Retention

Active Users

Subscriptions

Renewals

80. REVENUE ANALYTICS

Revenue By:

Plans

Ads

Featured Listings

Cities

Categories

Builders

Agencies

81. ADMIN PANEL

Admin Responsibilities:

Property Moderation

User Moderation

KYC Review

Support

Lead Monitoring

Reports

Content Moderation

82. PROPERTY MODERATION

Review:

New Listings

Edited Listings

Reported Listings

Duplicate Listings

Fraud Listings

Actions:

Approve

Reject

Suspend

Archive

83. USER MANAGEMENT

View Users

Suspend Users

Activate Users

Verify Users

Reset Access

Manage Roles

84. KYC MANAGEMENT

Review Documents

Approve

Reject

Request Resubmission

Track History

85. SUPPORT MANAGEMENT

Tickets

Escalations

Complaints

Fraud Reports

Abuse Reports

86. REPORTING SYSTEM

Reports:

Properties

Projects

Users

Leads

Revenue

Ads

Subscriptions

KYC

87. SUPER ADMIN CONTROL CENTER

Complete platform authority.

SUPER ADMIN MODULES

Users

Roles

Permissions

Plans

Payments

Advertisements

Feature Flags

Settings

Analytics

CMS

Support

Audit Logs

88. FEATURE FLAG SYSTEM

Enable / Disable:

AI Search

Voice Search

Ads

Payments

WhatsApp

Projects

Future Features

Without Deployment.

89. PLATFORM SETTINGS

Manage:

Branding

Logo

Colors

Domains

Email Settings

SMS Settings

WhatsApp Settings

Payment Settings

SEO Settings

90. ROLE MANAGEMENT

Roles:

Buyer/Seller

Broker

Builder

Agency

Admin

Super Admin

Permissions managed through RBAC.

91. AUDIT LOG SYSTEM

Track:

Login

Logout

Property Approval

Property Rejection

Payment Events

Role Changes

KYC Actions

Settings Changes

Admin Actions

Immutable logs.

92. FRAUD & COMPLIANCE CENTER

Monitor:

Spam Listings

Fake Accounts

Repeated OTP Abuse

Duplicate Properties

Suspicious Payments

Blacklisted Users

93. SUPPORT TICKET SYSTEM

Statuses:

Open

Assigned

In Progress

Waiting User

Resolved

Closed

94. KNOWLEDGE BASE

FAQs

Guides

Policies

Tutorials

Help Articles

95. PLATFORM HEALTH DASHBOARD

Monitor:

Server Health

Database Health

API Health

Payment Health

Notification Health

Search Health

Queue Health

96. PHASE 1 BUSINESS COMPLETION CRITERIA

Phase 1 complete only when:

✓ Property Module Complete

✓ Search Complete

✓ Lead CRM Complete

✓ Messaging Complete

✓ Subscription Complete

✓ Payment Complete

- ✓ Ads Complete
- ✓ KYC Complete
- ✓ Analytics Complete
- ✓ Admin Complete
- ✓ Security Complete
- ✓ Production Deployment Complete

PART 3 COMPLETE

MASTER BLUEPRINT v3.0 – PART 4

DATABASE ARCHITECTURE, RBAC, RLS, SECURITY, DEVOPS, SUPABASE, NEXT.JS, API ARCHITECTURE, INFRASTRUCTURE, DEPLOYMENT & PRODUCTION STANDARDS

Version: 3.0

Status: Production Ready Master Blueprint

97. FINAL TECHNOLOGY STACK

Frontend

Framework:

Next.js 15

Language:

TypeScript Strict Mode

Styling:

Tailwind CSS

UI Library:

ShadCN UI

Forms:

React Hook Form

Validation:

Zod

State Management:

Zustand

Icons:

Lucide

Charts:

Recharts

PWA:

Required

Backend

Platform:

Supabase

Components:

- PostgreSQL
 - Auth
 - Storage
 - Realtime
 - Edge Functions (Light Tasks Only)
-

Async Jobs

Primary:

Inngest

Alternative:

Trigger.dev

Heavy tasks must never execute directly inside user requests.

Infrastructure

CDN:

Cloudflare

Hosting:

Vercel

Alternative:

Hostinger VPS

Future:

Dedicated Cluster

98. DATABASE PRINCIPLES

Primary Keys:

UUID Only

No Auto Increment IDs

Mandatory Columns

Every Business Table Must Contain:

id

created_at

updated_at

created_by

updated_by

deleted_at

SOFT DELETE RULE

Never Hard Delete Business Data.

deleted_at = NULL

Means Active Record.

99. MULTI TENANT ARCHITECTURE

Every business table contains:

tenant_id

Examples:

Properties

Projects

Leads

Subscriptions

Payments

Advertisements

KYC

Notifications

Analytics

FUTURE TENANTS

Platform

Agency

Builder

Enterprise

Franchise

100. DATABASE MODULES

Authentication

Users

Roles

Permissions

Properties

Projects
Leads
Messaging
Subscriptions
Payments
Advertisements
KYC
Notifications
Analytics
CMS
Support
Audit Logs
Platform Settings
Feature Flags

101. DATABASE EXTENSIONS

Enable:

pgcrypto

uuid-oss

pg_trgm

btree_gin

citext

postgis

102. AUTHENTICATION ARCHITECTURE

Supported:

Email OTP

Mobile OTP

Google Login

FUTURE

Apple Login

Facebook Login

LinkedIn Login

SESSION MANAGEMENT

Device Tracking

Session Tracking

Login History

Security Logs

103. ROLE BASED ACCESS CONTROL

RBAC Mandatory.

ROLES

Buyer/Seller

Broker

Builder

Agency

Admin

Super Admin

PERMISSION FORMAT

module.action

Examples:

property.create

property.edit

property.delete

lead.view

lead.assign

payment.refund

cms.manage

settings.manage

104. PROPERTY PERMISSIONS

Create

Edit

Delete

Approve

Reject

Publish

Archive

Feature

Verify

Geo Verify

105. PROJECT PERMISSIONS

Create

Edit

Delete

Approve

Publish

Archive

106. LEAD PERMISSIONS

View

Create

Assign

Transfer

Export

Manage

107. PAYMENT PERMISSIONS

View

Refund

Verify

Generate Invoice

Manage

108. USER PERMISSIONS

View

Edit

Block

Unblock

Verify

Delete

109. ROW LEVEL SECURITY

Mandatory On Every Business Table.

No Exceptions.

RLS RULES

Users can access:

Own Data

Own Listings

Own Leads

Own Messages

Broker:

Own Leads

Assigned Leads

Agency:

Agency Data

Assigned Agent Data

Admin:

Moderation Access

Super Admin:

Global Access

110. AUDIT LOG ARCHITECTURE

Track:

Logins

Logouts

Property Changes

Project Changes

Payment Events

Role Changes

KYC Events

Admin Actions

Settings Changes

Feature Flag Changes

AUDIT DATA

User

Action

Module

Entity

Old Values

New Values

IP Address

Device

Timestamp

111. STORAGE ARCHITECTURE

Provider:

Supabase Storage

STORAGE BUCKETS

avatars

properties

projects

documents

brochures

floorplans

kyc

support

cms

temp

STORAGE RULES

Private By Default

RLS Protected

Virus Scan Required

Ownership Validation Required

112. IMAGE PROCESSING PIPELINE

Upload

↓

Virus Scan

↓

Duplicate Detection

↓

Compression

↓

Resize

↓

WebP

↓

AVIF

↓

Watermark

↓

Cloudflare CDN

↓

Storage

113. SEARCH DATABASE OPTIMIZATION

Indexes Required:

Property Type

City

State

Price

Status

Created Date

Listing Type

ADVANCED INDEXES

GIN

pg_trgm

PostGIS

Autocomplete Indexes

114. SEARCH PERFORMANCE TARGETS

Autocomplete:

< 200ms

Search:

< 1 Second

Property Detail:

< 500ms

Dashboard:

< 2 Seconds

115. API ARCHITECTURE

Pattern:

Controller

↓

Service

↓

Repository

↓

Database

API VERSIONING

/api/v1/

/api/v2/

Rules:

Never Break Existing APIs

Mobile Apps Must Support Versioning

Backward Compatibility Required

API RULES

Validation First

Authorization First

Rate Limiting

Error Logging

Audit Logging

116. API SECURITY

JWT Validation

Input Validation

Output Sanitization

SQL Injection Protection

XSS Protection

CSRF Protection

Rate Limiting

IP Monitoring

117. BACKGROUND WORKERS

Use:

Inngest

Trigger.dev

JOB TYPES

Notifications

Image Processing

Fraud Detection

Analytics Sync

Scheduled Reports

Plan Expiry

Subscription Renewal

118. NOTIFICATION INFRASTRUCTURE

Email

SMS

WhatsApp

Push

In-App

PROVIDERS

Email:

SMTP

SMS:

MSG91

WhatsApp:

Meta API

Push:

Firebase

119. FRAUD DETECTION ENGINE

Detect:

Duplicate Listings

Duplicate Images

Repeated OTP Requests

Spam Leads

Fake Accounts

Price Manipulation

Blacklisted Users

120. CMS ARCHITECTURE

Pages

Blogs

FAQs

Menus

Footer Links

SEO Metadata

Redirects

Schema Markup

121. SEO ARCHITECTURE

Server Side Rendering

Metadata

Schema.org

Open Graph

Twitter Cards

XML Sitemap

Robots.txt

Canonical URLs

122. ANALYTICS INFRASTRUCTURE

Primary:

PostHog

Secondary:

Google Analytics

EVENTS

Property View

Project View

Lead Generated

Call Click

WhatsApp Click

Visit Request

Search Query

Ad Click

Payment Success

Subscription Purchase

123. SECURITY MONITORING

Monitor:

Failed Logins

Permission Violations

RLS Violations

Suspicious Activity

Spam Activity

Fraud Signals

124. ENVIRONMENTS

Development

Staging

Production

RULES

Separate Databases

Separate Storage

Separate API Keys

Never Develop On Production

125. CI/CD PIPELINE

Developer Push

↓

GitHub

↓

Testing

↓

Build

↓

Deployment

↓

Monitoring

126. BACKUP STRATEGY

Database Backups

Storage Backups

Configuration Backups

Audit Backups

RETENTION

Daily

Weekly

Monthly

127. DISASTER RECOVERY

Backup Restore

Database Recovery

Storage Recovery

DNS Recovery

Rollback Support

128. PERFORMANCE TARGETS

Homepage:

< 2 Seconds

Search:

< 1 Second

Property:

< 500ms

API:

< 300ms

Lighthouse:

90+

Core Web Vitals:

Green

129. DEVELOPMENT ROADMAP

Phase 1

Foundation

Authentication

Properties

Projects

Search

Leads

Messaging

Subscriptions

Payments

Ads

KYC

Admin

Phase 2

Mobile Apps

Advanced AI

Advanced Analytics

Builder CRM

Agency CRM

Phase 3

National Expansion

Multi Language

Enterprise SaaS

API Marketplace

Partner Ecosystem

130. DEFINITION OF DONE

Platform is considered Production Ready only when:

- ✓ Authentication Complete
- ✓ RBAC Complete
- ✓ RLS Complete
- ✓ Property System Complete
- ✓ Project System Complete
- ✓ Search Complete
- ✓ Lead CRM Complete
- ✓ Messaging Complete
- ✓ Subscription Complete
- ✓ Payment Complete

- ✓ Advertisement Complete
 - ✓ KYC Complete
 - ✓ Analytics Complete
 - ✓ Admin Complete
 - ✓ Security Complete
 - ✓ SEO Complete
 - ✓ Monitoring Complete
 - ✓ Backup Complete
 - ✓ Deployment Complete
 - ✓ Documentation Complete
-

FINAL MASTER BLUEPRINT STATUS

Business Architecture:

COMPLETE

Product Architecture:

COMPLETE

Database Architecture:

COMPLETE

Security Architecture:

COMPLETE

Revenue Architecture:

COMPLETE

Frontend Architecture:

COMPLETE

Backend Architecture:

COMPLETE

DevOps Architecture:

COMPLETE

Deployment Architecture:

COMPLETE

Production Readiness:

YES

Claude / Gemini / ChatGPT Coding Ready:

YES

Master Blueprint v3.0:

COMPLETE

Part 4 Complete

MASTER BLUEPRINT v3.0 – PART 5

COMPLETE DATABASE MASTER SCHEMA & ENTITY ARCHITECTURE

Version: 3.0

Status: Developer Implementation Blueprint

131. DATABASE DESIGN PRINCIPLES

Database Engine:

PostgreSQL (Supabase)

Primary Key:

UUID

Soft Deletes:

deleted_at

Multi Tenant:

tenant_id

Audit Enabled:

Yes

RLS Enabled:

Mandatory

132. CORE AUTHENTICATION TABLES

users

Purpose:

Master user profile.

Fields:

- id
- tenant_id
- role_id
- first_name
- last_name
- display_name
- email
- mobile
- avatar_url
- city
- state
- country
- bio
- status
- email_verified
- mobile_verified
- last_login_at
- created_at
- updated_at
- deleted_at

user_sessions

Track:

- id
- user_id
- device
- browser
- ip_address
- login_at
- logout_at
- status

user_devices

Track:

- id
- user_id
- device_name
- device_type
- push_token

133. RBAC TABLES

roles

Examples:

- BuyerSeller
- Broker
- Builder
- Agency
- Admin

- SuperAdmin

permissions

Examples:

- property.create
- property.edit
- lead.assign
- payment.refund

role_permissions

Role mapping table.

user_roles

User role assignment.

134. TENANT TABLES

tenants

Examples:

- Platform
- Agency
- Builder Enterprise

Fields:

- id
- name
- type
- slug
- logo
- status

tenant_members

Tenant membership mapping.

135. PROPERTY TABLES

properties

Main property table.

Fields:

- id
- tenant_id
- owner_id
- property_type
- listing_type
- title
- description
- city
- locality
- latitude
- longitude
- price
- area
- status
- featured
- verified

property_details

Store:

- bhk

- bathrooms
 - balconies
 - furnishing
 - floor_number
 - total_floors
 - age
-

property_pricing

Store:

- base_price
 - rent
 - deposit
 - maintenance
-

property_media

Store:

- property_id
 - media_type
 - media_url
 - sort_order
-

property_amenities

Amenity mapping.

amenities

Master amenities.

property_documents

Store:

- floor plans
- ownership docs
- brochures

property_views

Track every property view.

property_favorites

Saved properties.

property_comparisons

Comparison tracking.

property_reports

User reported listings.

property_status_logs

Track status history.

136. PROJECT TABLES

projects

Main project.

project_phases

Project phases.

project_towers

Tower details.

project_units

Inventory units.

project_media

Project media.

project_brochures

Project brochures.

project_updates

Construction updates.

project_amenities

Project amenities.

137. SEARCH TABLES

search_logs

Track searches.

saved_searches

User saved searches.

search_alerts

Matching alerts.

search_synonyms

Examples:

Ghar → Residential

Shop → Commercial

138. LEAD CRM TABLES

leads

Master lead table.

Fields:

- source
- status
- assigned_to
- property_id
- project_id

lead_notes

CRM notes.

lead_followups

Follow-ups.

lead_activities

All lead actions.

lead_assignments

Assignment history.

lead_sources

Master sources.

lead_status_history

Lead lifecycle tracking.

139. VISIT MANAGEMENT TABLES

visit_requests

Visit bookings.

visit_feedback

Visit feedback.

visit_status_history

Visit tracking.

140. CHAT TABLES

conversations

Chat room.

conversation_participants

Participants.

messages

Messages.

message_attachments

Media files.

message_reads

Read tracking.

141. SUBSCRIPTION TABLES

plans

Master plans.

plan_features

Plan limits.

subscriptions

Active subscriptions.

subscription_history

History.

plan_usage

Usage counters.

142. PAYMENT TABLES

payments

Transactions.

payment_webhooks

Webhook logs.

invoices

Invoices.

refunds

Refund records.

payment_reconciliation

Gateway reconciliation.

143. ADVERTISEMENT TABLES

ad_campaigns

Campaigns.

ad_packages

Packages.

advertisements

Advertisements.

ad_impressions

Impression logs.

ad_clicks

Click logs.

featured_listings

Featured placements.

144. KYC TABLES

kyc_profiles

Verification profile.

kyc_documents

Documents.

kyc_reviews

Review history.

verification_badges

Badge assignments.

145. NOTIFICATION TABLES

notifications

Master notifications.

notification_templates

Email/SMS templates.

notification_logs

Delivery logs.

notification_preferences

User preferences.

146. ANALYTICS TABLES

analytics_events

Event tracking.

property_analytics

Property metrics.

project_analytics

Project metrics.

revenue_analytics

Revenue metrics.

user_analytics

User metrics.

147. SUPPORT TABLES

support_tickets

Tickets.

support_messages

Ticket chat.

support_categories

Categories.

support_assignments

Assignment history.

148. CMS TABLES

pages

CMS pages.

blogs

Blog articles.

blog_categories

Categories.

faqs

FAQs.

menus

Navigation.

149. PLATFORM SETTINGS TABLES

platform_settings

Global settings.

feature_flags

Feature toggles.

integrations

External services.

150. SECURITY TABLES

audit_logs

Immutable logs.

security_logs

Security events.

login_attempts

Failed logins.

blocked_ips

Blocked IPs.

fraud_flags

Fraud monitoring.

151. DATABASE INDEX STRATEGY

Mandatory Indexes:

Properties:

- city
- locality
- property_type
- listing_type
- price
- status

Leads:

- assigned_to
- status

Users:

- email
- mobile

Payments:

- payment_id
- status

Search:

- pg_trgm
- GIN
- PostGIS

152. ESTIMATED TABLE COUNT

Authentication: 4

RBAC: 4

Tenants: 2

Properties: 12+

Projects: 8+

Search: 4

CRM: 6+

Visits: 3

Messaging: 5

Subscriptions: 4

Payments: 5

Ads: 5

KYC: 4

Notifications: 4

Analytics: 4

Support: 4

CMS: 5

Settings: 3

Security: 5

Total:

70–90 Core Tables

Enterprise Expansion:

100–120+ Tables

PART 5 COMPLETE

MASTER BLUEPRINT v3.0 – PART 6

COMPLETE FRONTEND ARCHITECTURE, PUBLIC WEBSITE, DASHBOARDS,
NAVIGATION, UI SYSTEM & USER EXPERIENCE

Version: 3.0

Status: Frontend Implementation Blueprint

153. FRONTEND ARCHITECTURE OVERVIEW

Frontend Framework:

Next.js 15

Architecture:

App Router

Server Components First

Client Components Only When Required

DESIGN PRINCIPLES

Mobile First

Fast Loading

SEO First

Accessibility Ready

Scalable Components

Reusable UI

154. WEBSITE STRUCTURE

Public Website

↓

Authentication

↓

User Dashboard

↓

Role Dashboard

↓

Admin

↓

Super Admin

155. PUBLIC WEBSITE PAGES

Homepage

Property Search

Property Listing

Property Details

Project Listing

Project Details

Builders Directory

Brokers Directory

Agencies Directory

Blog

FAQ

Contact Us

About Us

Privacy Policy

Terms & Conditions

Refund Policy

Careers

Support

156. HOMEPAGE STRUCTURE

SECTION 1

Header

Logo

Search Bar

Login

Register

Post Property

SECTION 2

Hero Search

Location Search

Property Type

Budget

Search Button

Voice Search

AI Search

SECTION 3

Featured Properties

SECTION 4

Latest Properties

SECTION 5

Premium Projects

SECTION 6

Top Builders

SECTION 7

Top Brokers

SECTION 8

Popular Cities

SECTION 9

Property Categories

Residential

Commercial

Plots

Industrial

Agricultural

Rental

SECTION 10

Recently Added

SECTION 11

Price Drop Properties

SECTION 12

Blogs

SECTION 13

FAQs

SECTION 14

Footer

157. HEADER ARCHITECTURE

Logo

Search

Properties

Projects

Builders

Brokers

Agencies

Pricing

Blog

Support

Login

Register

Post Property

158. FOOTER ARCHITECTURE

[Company](#)

[About](#)

[Contact](#)

[Careers](#)

[Support](#)

[Legal](#)

[Privacy](#)

[Terms](#)

[Refund](#)

[Resources](#)

[Blog](#)

[Guides](#)

[FAQs](#)

[Cities](#)

[Ahmedabad](#)

[Rajkot](#)

[Surat](#)

[Vadodara](#)

[Jamnagar](#)

[More](#)

159. PROPERTY SEARCH PAGE

Components:

[Search Bar](#)

[Map View](#)

[List View](#)

[Filters](#)

[Sorting](#)

Pagination

FILTER PANEL

Location

Property Type

Listing Type

Budget

Area

BHK

Amenities

Builder

RERA

Vastu

Geo Verified

Furnishing

Property Age

160. PROPERTY DETAIL PAGE

Property Gallery

Property Summary

Price

Amenities

Location Map

Description

Floor Plans

Documents

Nearby Places

Builder Information

Owner Information

Lead Form

WhatsApp

Call Button

Book Visit

Report Property

Save Property

Compare Property

Share Property

Similar Properties

161. PROJECT LISTING PAGE

Search

Filters

Project Cards

Builder Filter

Possession Filter

Budget Filter

RERA Filter

162. PROJECT DETAIL PAGE

Project Overview

Builder Information

Amenities

Tower Information

Inventory

Floor Plans

Brochures

Construction Updates

Location Map

Lead Form

Book Site Visit

163. AUTHENTICATION PAGES

Login

Register

Verify OTP

Forgot Password

Reset Password

Complete Profile

164. USER ONBOARDING FLOW UI

Select Role

Buyer/Seller

Broker

Builder

Agency

↓

Verify Mobile

↓

Verify Email

↓

Complete Profile

↓

Dashboard

165. BUYER/SELLER DASHBOARD

Dashboard Home

My Profile

Saved Properties

Compared Properties

Property Alerts

Visit Requests

Messages

Notifications

My Listings

My Leads

Subscription

Payments

Settings

DASHBOARD WIDGETS

Saved Properties

Recent Searches

Latest Messages

Visit Requests

Lead Summary

Notifications

166. BROKER DASHBOARD

Overview

Properties

Leads

Lead CRM

Visits

Messages

Analytics

Advertisements

Subscription

Payments

Reports

Settings

BROKER WIDGETS

Active Listings

New Leads

Today's Follow Ups

Visits Scheduled

Revenue

Conversion Rate

167. BUILDER DASHBOARD

Overview

Projects

Inventory

Construction Updates

Leads

Site Visits

Analytics

Advertisements

Subscription

Payments

Settings

BUILDER WIDGETS

Projects

Available Units

Booked Units

Lead Count

Visit Requests

Revenue

168. AGENCY DASHBOARD

Overview

Agents

Leads

Properties

Projects

CRM

Assignments

Analytics

Advertisements

Subscription

Reports

Settings

AGENCY WIDGETS

Total Agents

Assigned Leads

Open Leads

Conversion Rate

Revenue

Team Performance

169. LEAD CRM INTERFACE

Lead List

Kanban Board

Calendar View

Follow Ups

Lead Details

Communication History

Assignments

Status Changes

170. CHAT INTERFACE

Conversation List

Property Chat

Project Chat

Message Area

Attachments

Unread Count

Notifications

171. VISIT MANAGEMENT UI

Visit Requests

Calendar

Upcoming Visits

Completed Visits

Feedback

Status Tracking

172. SUBSCRIPTION UI

Current Plan

Plan Usage

Upgrade Plan

Renew Plan

Invoices

Payment History

173. ADVERTISEMENT UI

Campaigns

Create Campaign

Performance

Impressions

Clicks

Leads

Budget

174. PROFILE MANAGEMENT UI

Personal Details

Business Details

KYC

Documents

Social Links

Notifications

Security

175. ADMIN DASHBOARD

Overview

Users

Properties

Projects

Leads

KYC

Payments

Ads

CMS

Support

Analytics

Audit Logs

Settings

176. ADMIN PROPERTY MODERATION

Pending Listings

Approved Listings

Rejected Listings

Reported Listings

Fraud Listings

Actions:

Approve

Reject

Suspend

Archive

177. ADMIN KYC MODULE

Pending KYC

Approved KYC

Rejected KYC

Review Documents

Verification History

178. SUPPORT DESK UI

Tickets

Assignments

Responses

Escalations

Resolved

Closed

179. CMS UI

Pages

Blogs

FAQs

Menus

SEO

Media Library

180. SUPER ADMIN DASHBOARD

Platform Overview

Users

Roles

Permissions

Tenants

Plans

Payments

Ads

Feature Flags

Integrations

Analytics

Audit Logs

Settings

181. SUPER ADMIN SETTINGS

Branding

Domains

Email

SMS

WhatsApp

Payment Gateways

Storage

CDN

SEO

Security

182. NOTIFICATION CENTER UI

Unread

Read

Archived

System Notifications

Lead Notifications

Payment Notifications

Visit Notifications

183. GLOBAL SEARCH COMPONENT

Search:

Properties

Projects

Users

Builders

Brokers

Agencies

Blogs

184. REUSABLE UI COMPONENTS

Button

Input

Select

Dropdown

Modal

Drawer

Tooltip

Tabs

Accordion

Card

Badge

Avatar

Pagination

Data Table

Charts

Calendar

Kanban

Uploader

Map

185. MOBILE EXPERIENCE

Bottom Navigation

Search

Saved

Messages

Notifications

Profile

186. RESPONSIVE BREAKPOINTS

Mobile

Tablet

Laptop

Desktop

Large Desktop

187. ACCESSIBILITY

Keyboard Navigation

Screen Reader Support

ARIA Labels

Color Contrast

Focus States

188. DESIGN SYSTEM

Typography

Spacing

Colors

Icons

Buttons

Forms

Cards

Tables

Charts

Badges

189. ERROR PAGES

401 Unauthorized

403 Forbidden

404 Not Found

429 Too Many Requests

500 Server Error

Maintenance Page

190. LOADING STATES

Skeleton Loaders

Progress Bars

Lazy Loading

Infinite Scroll

Image Placeholders

191. EMPTY STATES

No Properties

No Leads

No Messages

No Projects

No Notifications

No Results

192. FRONTEND PERFORMANCE RULES

Code Splitting

Lazy Loading

Image Optimization

Caching

SSR

SEO Optimization

193. PWA FEATURES

Install App

Offline Mode

Push Notifications

Home Screen Shortcut

194. MULTI LANGUAGE READY

English

Hindi

Gujarati

Future Expansion Supported

195. FRONTEND COMPLETION CRITERIA

- ✓ Homepage Complete
- ✓ Property Pages Complete
- ✓ Project Pages Complete
- ✓ Search Complete
- ✓ Authentication Complete
- ✓ Buyer Dashboard Complete
- ✓ Broker Dashboard Complete
- ✓ Builder Dashboard Complete
- ✓ Agency Dashboard Complete
- ✓ Admin Dashboard Complete
- ✓ Super Admin Dashboard Complete
- ✓ Mobile Responsive Complete
- ✓ Accessibility Complete
- ✓ SEO Complete

PART 6 COMPLETE

MASTER BLUEPRINT v3.0 – PART 7

COMPLETE BACKEND ARCHITECTURE, API SPECIFICATIONS, SERVICES, REPOSITORIES, WEBHOOKS, QUEUE WORKERS, SEARCH ENGINE INTERNALS & INTEGRATION ARCHITECTURE

Version: 3.0

Status: Backend Implementation Blueprint

196. BACKEND ARCHITECTURE OVERVIEW

Architecture Pattern:

Client

↓

API Layer

↓

Controller Layer

↓

Service Layer

↓

Repository Layer

↓

Database Layer

↓

Supabase PostgreSQL

PRINCIPLES

Thin Controllers

Fat Services

Reusable Repositories

Strict Validation

Audit Logging

RLS Enforcement

Multi Tenant Support

197. APPLICATION LAYERS

Presentation Layer

API Routes

Webhooks

Realtime Events

Business Layer

Services

Business Rules

Workflow Logic

Data Layer

Repositories

Database Queries

Storage Access

Infrastructure Layer

Supabase

Redis

Inngest

Cloudflare

198. API VERSIONING

Supported:

/api/v1

/api/v2

Future:

/api/v3

RULES

Never Break Existing APIs

Maintain Backward Compatibility

Version Every Public Endpoint

199. AUTHENTICATION APIs

POST

/auth/register

/auth/login

/auth/logout

/auth/refresh

/auth/verify-otp

/auth/forgot-password

/auth/reset-password

/auth/google-login

200. USER APIs

GET

/users/me

/users/profile

/users/settings

POST

/users/update

/users/avatar

/users/preferences

201. PROPERTY APIs

POST

/properties/create

PUT

/properties/update

DELETE

/properties/delete

GET

/properties

/properties/{id}

/properties/search

/properties/featured

/properties/recommended

/properties/nearby

/properties/price-drop

202. PROPERTY ACTION APIs

POST

/properties/save

/properties/compare

/properties/report

/properties/share

/properties/contact

/properties/book-visit

203. PROJECT APIs

GET

/projects

/projects/{id}

POST

/projects/create

/projects/update

DELETE

/projects/delete

204. SEARCH APIs

GET

/search

/search/properties

/search/projects

/search/builders

/search/brokers

/search/agencies

POST

/search/ai

/search/voice

205. LEAD APIs

GET

/leads

/leads/{id}

POST

/leads/create

/leads/update

/leads/assign

/leads/transfer

/leads/close

206. VISIT APIs

POST

/visits/request

/visits/approve

/visits/reject

/visits/reschedule

GET

/visits

/visits/calendar

207. CHAT APIs

GET

/conversations

/messages

POST

/messages/send

/messages/read

/messages/upload

208. SUBSCRIPTION APIs

GET

/plans

/subscriptions

POST

/subscriptions/create

/subscriptions/renew

/subscriptions/cancel

/subscriptions/upgrade

209. PAYMENT APIs

POST

/payments/create-order

/payments/verify

/payments/refund

GET

/payments/history

/invoices

210. ADVERTISEMENT APIs

GET

/ads

/campaigns

POST

/campaigns/create

/campaigns/update

/campaigns/pause

/campaigns/delete

211. KYC APIs

POST

/kyc/submit

/kyc/update

GET

/kyc/status

/kyc/documents

212. NOTIFICATION APIs

GET

/notifications

POST

/notifications/read

/notifications/archive

213. ADMIN APIs

Users

Properties

Projects

Leads

Payments

KYC

Support

CMS

Analytics

Settings

Audit Logs

214. CONTROLLER RESPONSIBILITIES

Controllers Only:

Validate Request

Authorize User

Call Service

Return Response

Controllers Must Never:

Contain Business Logic

Contain SQL Queries

Access Database Directly

215. SERVICE LAYER RESPONSIBILITIES

Services Handle:

Business Rules

Workflows

Permissions

Validation

Notifications

Audit Events

Queue Events

216. REPOSITORY LAYER

Responsibilities:

Database Access

Data Retrieval

Filtering

Pagination

Search Queries

Transactions

Repositories Must Not:

Contain Business Logic

217. VALIDATION LAYER

Framework:

Zod

Validation Types:

Request Validation

Response Validation

Environment Validation

Webhook Validation

218. ERROR HANDLING

Standard Response:

success

message

data

errors

request_id

timestamp

ERROR CODES

400

401

403

404

409

422

429

500

219. PAGINATION STANDARD

Response:

page

limit

total

total_pages

items

220. FILTERING STANDARD

Support:

Location

Price

Area

Property Type

Status

Builder

Agency

Broker

Date

221. SORTING STANDARD

Newest

Oldest

Price Low To High

Price High To Low

Most Viewed

Featured First

222. FILE UPLOAD SERVICE

Supported:

Images

PDF

Documents

Brochures

Floor Plans

Videos

VALIDATIONS

File Size

File Type

Virus Scan

Duplicate Detection

Ownership Check

223. STORAGE SERVICE

Upload

Delete

Replace

Download

Secure Access

Signed URLs

224. IMAGE SERVICE

Tasks:

Compression

Resize

Crop

Watermark

WebP

AVIF

Thumbnail Generation

225. SEARCH ENGINE INTERNALS

Phase 1:

PostgreSQL

pg_trgm

GIN Indexes

PostGIS

Phase 2:

Redis Cache

Phase 3:

OpenSearch

226. SEARCH PIPELINE

Query

↓

Normalize

↓

Synonym Match

↓

Filter Extraction

↓

Database Search

↓

Ranking

↓

Response

227. AI SEARCH SERVICE

Input:

Natural Language

↓

Intent Detection

↓

Location Extraction

↓

Budget Extraction

↓

Property Type Extraction

↓

Search Filters

↓

Database Query

228. SEARCH CACHE

Cache:

Popular Searches

City Searches

Featured Properties

Homepage Data

Provider:

Redis

Upstash

229. LEAD ROUTING SERVICE

Logic:

Property Owner

↓

Assigned Broker

↓

Agency

↓

Admin Queue

Round Robin Supported

230. CRM AUTOMATION SERVICE

Auto Assign

Auto Follow Up

Lead Scoring

Lead Aging

Reminder Generation

231. NOTIFICATION ENGINE

Triggers:

Lead

Visit

Payment

Message

Property

Project

Subscription

232. EMAIL SERVICE

Templates

Variables

Tracking

Retries

Logs

233. SMS SERVICE

Provider:

MSG91

Features:

OTP

Alerts

Notifications

Tracking

234. WHATSAPP SERVICE

Provider:

Meta API

Features:

OTP

Notifications

Lead Alerts

Visit Alerts

Payment Alerts

235. PUSH NOTIFICATION SERVICE

Provider:

Firebase

Features:

Realtime Delivery

Deep Linking

Analytics

236. WEBHOOK ARCHITECTURE

Providers:

Razorpay

MSG91

Meta

Custom Systems

237. WEBHOOK SECURITY

Validate Signature

Verify Source

Prevent Replay Attacks

Log Every Request

238. WEBHOOK EVENTS

Payment Success

Payment Failed

Refund

Subscription Renewed

Subscription Expired

239. QUEUE ARCHITECTURE

Provider:

Inngest

Alternative:

Trigger.dev

240. QUEUE TYPES

Email Queue

SMS Queue

WhatsApp Queue

Image Queue

Analytics Queue

Fraud Queue

Search Queue

241. IMAGE PROCESSING WORKER

Upload

↓

Virus Scan

↓

Duplicate Check

↓

Compress

↓

WebP

↓

AVIF

↓

Watermark

↓

Storage

242. FRAUD DETECTION WORKER

Check:

Duplicate Listings

Duplicate Images

Duplicate Numbers

Spam Leads

Blacklisted Users

243. ANALYTICS WORKER

Collect Events

Aggregate Data

Store Metrics

Generate Reports

244. SUBSCRIPTION WORKER

Check Expiry

Send Alerts

Renew Plans

Deactivate Plans

Update Listings

245. AUDIT WORKER

Track:

Logins

Payments

Role Changes

Property Updates

Settings Changes

246. INTEGRATION ARCHITECTURE

External Systems:

Razorpay

MSG91

Meta WhatsApp

Firebase

Google Analytics

PostHog

Cloudflare

Ola Maps

Google Maps

247. THIRD PARTY CONFIGURATION

All Credentials Stored In:

Environment Variables

Never Hardcoded

248. RATE LIMITING

Authentication APIs

OTP APIs

Search APIs

Contact APIs

Lead APIs

Webhook APIs

249. SECURITY MIDDLEWARE

Authentication

Authorization

Rate Limiting

Request Logging

Audit Logging

Tenant Isolation

250. BACKEND MONITORING

Monitor:

API Response Time

Queue Health

Webhook Failures

Database Health

Storage Health

Search Health

251. LOGGING SYSTEM

Application Logs

Security Logs

Audit Logs

Payment Logs

Notification Logs

Error Logs

252. BACKEND TESTING

Unit Tests

Integration Tests

API Tests

Security Tests

Performance Tests

253. BACKEND COMPLETION CRITERIA

✓ APIs Complete

✓ Services Complete

✓ Repositories Complete

✓ Search Engine Complete

✓ Notifications Complete

✓ Webhooks Complete

✓ Queue Workers Complete

✓ Security Complete

✓ Monitoring Complete

✓ Testing Complete

PART 7 COMPLETE

MASTER BLUEPRINT v3.0 – PART 8

FINAL IMPLEMENTATION ROADMAP, FOLDER STRUCTURE, DEVELOPMENT PHASES,
TESTING, CI/CD, SOPs, GO-LIVE CHECKLIST & PRODUCTION LAUNCH BLUEPRINT

Version: 3.0

Status: Final Production Launch Blueprint

254. PROJECT EXECUTION STRATEGY

Development Approach:

Phase Based

Feature Driven

Module Based

Production First

PRIORITY ORDER

1. Foundation
2. Authentication
3. Properties
4. Search
5. Leads
6. Messaging
7. Subscription
8. Payments
9. Ads
10. KYC
11. Analytics
12. Admin

13. Production

255. FINAL FOLDER STRUCTURE

realestate-platform/

src/

app/

(auth)

(properties)

(projects)

(search)

(messages)

(dashboard)

(admin)

api/

components/

features/

property/

project/

lead/

search/

messaging/

subscription/

payment/

advertisement/

analytics/

stores/

hooks/

services/

repositories/

validators/

types/

constants/

lib/

utils/

middleware/

supabase/

migrations/

seed/

workers/

inngest/

cron/

tests/

unit/

integration/

e2e/

docs/

architecture/

api/

database/

deployment/

256. DEVELOPMENT ENVIRONMENTS

Development

Local Developer Environment

Staging

QA Testing Environment

Production

Live Environment

Rules:

Separate Database

Separate Storage

Separate Secrets

Separate Domains

257. GIT STRATEGY

Main Branch

production

Develop Branch

development

Feature Branches

feature/property

feature/search

feature/payment

feature/crm

Bug Branches

fix/property

fix/payment

258. CODE REVIEW RULES

Mandatory Before Merge:

Code Review

Security Review

Performance Review

Testing Passed

Build Passed

259. DEVELOPMENT PHASE 1

Foundation

Authentication

RBAC

RLS

Users

Profiles

Settings

Expected Output:

Secure Platform Base

260. DEVELOPMENT PHASE 2

Property System

Property Media

Property Search

Property Details

Maps

Amenities

Saved Properties

Compare Properties

Expected Output:

Working Property Marketplace

261. DEVELOPMENT PHASE 3

Projects

Builders

Inventory

Project Leads

Site Visits

Construction Updates

Expected Output:

Builder Platform

262. DEVELOPMENT PHASE 4

CRM

Lead Management

Lead Routing

Follow Ups

Visits

Reminders

Messaging

Expected Output:

Complete Lead Engine

263. DEVELOPMENT PHASE 5

Subscriptions

Plans

Payments

Invoices

Refunds

Usage Tracking

Expected Output:

Monetization Ready

264. DEVELOPMENT PHASE 6

Advertisements

Featured Listings

Campaigns

Analytics

Tracking

Expected Output:

Advertising Platform

265. DEVELOPMENT PHASE 7

KYC

Verification

Moderation

Fraud Detection

Support

Expected Output:

Trust & Compliance

266. DEVELOPMENT PHASE 8

Admin

Super Admin

CMS

Settings

Feature Flags

Audit Logs

Expected Output:

Platform Control Center

267. DEVELOPMENT PHASE 9

Performance

Caching

Optimization

Security Hardening

Load Testing

Expected Output:

Production Ready

268. TESTING STRATEGY

Unit Testing

Integration Testing

API Testing

UI Testing

Security Testing

Performance Testing

Load Testing

UAT Testing

269. UNIT TEST COVERAGE

Services

Validators

Repositories

Utilities

Permissions

Target:

80%+

Coverage

270. API TESTING

Authentication APIs

Property APIs

Lead APIs

Payment APIs

Search APIs

Messaging APIs

271. SECURITY TESTING

JWT

RBAC

RLS

XSS

CSRF

SQL Injection

Rate Limits

File Upload Security

272. PERFORMANCE TESTING

Homepage

Search

Property Detail

Lead Creation

Messaging

Payments

Targets:

API < 300ms

Search < 1s

Property < 500ms

273. USER ACCEPTANCE TESTING

Buyer

Seller

Broker

Builder

Agency

Admin

Super Admin

All User Journeys Tested

274. CI/CD PIPELINE

Git Push

↓

GitHub

↓

Lint

↓

Type Check

↓

Tests

↓

Build

↓

Security Scan

↓

Deploy

↓

Monitoring

275. BUILD VALIDATION

No Type Errors

No Build Errors

No Security Errors

No Failed Tests

276. MONITORING STACK

Application Monitoring

Database Monitoring

API Monitoring

Queue Monitoring

Storage Monitoring

Search Monitoring

Tools:

PostHog

Supabase

Cloudflare

Custom Logs

277. LOGGING STRATEGY

Application Logs

Error Logs

Security Logs

Audit Logs

Payment Logs

Notification Logs

Search Logs

278. BACKUP STRATEGY

Database

Storage

Configurations

Audit Logs

CMS Data

Schedule:

Daily

Weekly

Monthly

279. DISASTER RECOVERY PLAN

Database Restore

Storage Restore

Rollback Deployment

DNS Recovery

Infrastructure Recovery

280. SEO LAUNCH CHECKLIST

Metadata

Schema

Canonical URLs

Sitemap

Robots.txt

Open Graph

Twitter Cards

281. SECURITY LAUNCH CHECKLIST

RBAC Verified

RLS Verified

JWT Verified

Rate Limiting Active

File Upload Security Active

Audit Logs Active

282. BUSINESS LAUNCH CHECKLIST

Plans Created

Payments Active

Ads Active

KYC Active

CRM Active

Notifications Active

Support Active

283. CONTENT LAUNCH CHECKLIST

Homepage Content

Blogs

FAQs

Policies

Support Articles

City Pages

Builder Pages

284. GO-LIVE CHECKLIST

Production Domain

SSL

CDN

Monitoring

Backups

Payment Gateway

SMS Gateway

WhatsApp Gateway

Push Notifications

Analytics

SEO

285. POST LAUNCH SOP

Daily

Monitor Errors

Monitor Payments

Monitor Leads

Monitor Support

Weekly

Performance Review

Revenue Review

Security Review

Fraud Review

Monthly

Growth Review

Infrastructure Review

Roadmap Review

286. CUSTOMER SUPPORT SOP

Ticket Creation

Assignment

Response

Escalation

Resolution

Closure

Feedback

287. REVENUE OPERATIONS SOP

Monitor:

Subscriptions

Payments

Refunds

Ads

Invoices

Taxes

288. FRAUD OPERATIONS SOP

Review:

Duplicate Listings

Spam Leads

Fake Accounts

Abuse Reports

Blacklisted Users

289. SCALING ROADMAP

Phase 1

Gujarat

Phase 2

Major Indian Cities

Phase 3

Pan India

Phase 4

Enterprise SaaS

Phase 5

International Expansion

290. FUTURE MODULES

Mobile Apps

AI Assistant

AI Recommendations

AI Pricing

AI Lead Scoring

AI Property Matching

Voice Assistant

Partner APIs

White Label SaaS

Franchise Management

291. FINAL PRODUCTION ACCEPTANCE CRITERIA

Authentication Complete

RBAC Complete

RLS Complete
Property System Complete
Project System Complete
Search Complete
CRM Complete
Messaging Complete
Subscription Complete
Payments Complete
Advertisements Complete
KYC Complete
Analytics Complete
Admin Complete
Security Complete
CMS Complete
Monitoring Complete
Backups Complete
Disaster Recovery Complete
Documentation Complete

292. FINAL PLATFORM STATUS

Business Architecture:

COMPLETE

Product Architecture:

COMPLETE

Frontend Architecture:

COMPLETE

Backend Architecture:

COMPLETE

Database Architecture:

COMPLETE

Security Architecture:

COMPLETE

Revenue Architecture:

COMPLETE

DevOps Architecture:

COMPLETE

Operations Architecture:

COMPLETE

Launch Blueprint:

COMPLETE

FINAL MASTER BLUEPRINT v3.0 STATUS

Part 1:

Business Foundation

COMPLETE

Part 2:

Property, Search, CRM

COMPLETE

Part 3:

Revenue, Ads, Payments

COMPLETE

Part 4:

Infrastructure & Security

COMPLETE

Part 5:

Database Master Schema

COMPLETE

Part 6:

Frontend & Dashboards

COMPLETE

Part 7:

Backend & APIs

COMPLETE

Part 8:

Implementation & Launch

COMPLETE

MASTER BLUEPRINT v3.0

STATUS:

100% COMPLETE

READY FOR:

✓ Claude Code

✓ Gemini

✓ ChatGPT

✓ Cursor

✓ Windsurf

✓ Development Team

✓ Production Planning

END OF MASTER BLUEPRINT